Dashbite — Core Data Model & API Contract Pack (v1.0)

Date: November 1, 2026
Owner: Cyrel (PM)
Approved By: Cyrel
Status: Approved for Week 2 Development

1. The Status Vocabulary (The Dictionary)

Rule: Frontend and Backend MUST use these exact words. No synonyms allowed.

Order Statuses:

- PENDING: Order placed, waiting for restaurant review.
- ACCEPTED: Restaurant accepted, food is being prepared.
- REJECTED: Restaurant rejected the order.
- SEARCHING_RIDER: System is doing the 1km blast.
- NO_RIDER: Service boundary reached, no rider accepted.
- PICKED_UP: Rider has the food.
- DELIVERED: Order handed to customer.
- CANCELLED: Order cancelled by customer or Admin.

Account Statuses:

- APPROVED: Full access to role features.
- PENDING_APPROVAL: Can log in, but operational actions are blocked (Riders/Restaurants only).
- DISABLED: Cannot log in.

Payment Statuses (COD & Online):

- UNPAID: Awaiting payment.
- COLLECTED_BY_RIDER: COD cash is in the rider's hands.
- REMITTED_TO_ADMIN: Rider handed cash to Admin.
- SETTLED_TO_RESTAURANT: Admin paid the restaurant.
- FLAGGED_DISCREPANCY: Short remittance or mismatch (goes to Admin queue).

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2. Core Data Models (The Entities)

Rule: These are the required fields the database must store and the frontend will send/receive.

A. User Model

- id (UUID)

- role (ENUM: CUSTOMER, RESTAURANT, RIDER, ADMIN)
- email (String)
- account_status (ENUM: APPROVED, PENDING_APPROVAL, DISABLED)

B. Order Model

- id (UUID)
- customer_id (UUID)
- restaurant_id (UUID)
- rider_id (UUID, nullable)
- order_status (ENUM from section 1)
- food_subtotal (Decimal)
- delivery_fee (Decimal)
- total_amount (Decimal)
- delivery_address (String)
- payment_method (ENUM: COD, ONLINE)

C. Dispatch/Rider Offer Model

- offer_id (UUID)
- order_id (UUID)
- rider_id (UUID)
- ring_radius_km (Integer: starts at 1, increments by 1)
- offer_status (ENUM: BLASTED, ACCEPTED, EXPIRED, CLOSED)

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3. API Endpoints & Error Outcomes

Rule: Every API call must have a defined success and failure response.

Endpoint 1: User Login

- POST /api/auth/login
- **Success (200 OK):** Returns { "token": "jwt...", "role": "RIDER", "status": "PENDING_APPROVAL" }
- **Error (401 Unauthorized):** { "error": "INVALID_CREDENTIALS", "message": "Login failed." }
- **Error (429 Too Many Requests):** { "error": "OTP_THROTTLED", "message": "Too many requests. Try again in 60 seconds." }

Endpoint 2: Create Order (Customer)

- POST /api/orders

- **Success (201 Created):** { "order_id": "123", "status": "PENDING" }
- **Error (400 Bad Request):** { "error": "OUT_OF_STOCK", "message": "Item 'Burger' is no longer available." }
- **Error (400 Bad Request):** { "error": "OUT_OF_BOUNDARY", "message": "Delivery address is outside the service area." }

Endpoint 3: Rider Atomic Claim (The Race Condition)

- POST /api/dispatch/accept-offer
- **Payload:** { "order_id": "123", "rider_id": "456" }
- **Success (200 OK - Winner):** { "status": "ASSIGNED", "message": "You have accepted the order." }
- **Error (409 Conflict - Loser):** { "error": "OFFER_ALREADY_CLAIMED", "message": "Offer closed. Another rider accepted it first." }

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4. Role Permissions Matrix (Who can do what)

Rule: Prevent unauthorized access (REQ-08).

Action	Customer	Restaurant	Rider	Admin
Create Order	✅	✅ (External)	❌	❌
Accept/Reject Order	❌	✅	❌	❌
Blast Offer	❌	❌	❌	System Only
Claim Offer	❌	❌	✅	❌
Change Delivery Fee	❌	❌	❌	✅
Approve Accounts	❌	❌	❌	✅

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5. Version Control Rule

- Any change to these fields, endpoints, or status words requires a **v1.1 update** to this document.
- Cyrel (PM) and Jelvie (Backend) must sign off on changes before frontend repos (Archie) update their code.